

# Projectile motion: Answers

Pearson Edexcel AS Level Physics · 4 questions · 23 marks

**1** Physics · Specimen 2015 · Paper 01 · Q15

(a) Use of  $v^2 = u^2 + 2as$  (1) OR use of  $\frac{1}{2}mv^2 = mgh$  (1) (2)

Initial speed =  $7.0 \text{ m s}^{-1}$  (1)

Example of calculation:  $v = 0$ ,  $a = -9.81 \text{ m s}^{-2}$ ,  $s = 2.5 \text{ m}$ ;  $u^2 = -2as = 2 \times 9.81 \times 2.5 = 49 \text{ m}^2 \text{ s}^{-2}$ ;  $u = 7.0 \text{ m s}^{-1}$

Alternative:  $v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.5} = 7.0 \text{ m s}^{-1}$

(b) Use of trig function to find  $v$  vertical (1) (5)

Use of trig function to find  $v$  horizontal (1)

Use of equation of motion to find time of flight (1)

Use of equation of motion to find distance (1)

Horizontal distance =  $2.7 \text{ m}$  (1)

Example of calculation (as printed in the mark scheme): vertical velocity =  $6.5 \text{ m s}^{-1} \sin 20 = 2.22 \text{ m s}^{-1}$ ; time of flight using  $v = u + at$ :  $-2.22 = 2.22 + (-9.81 \times t)$ ,  $t = 0.45 \text{ s}$ ;

horizontal velocity =  $6.5 \text{ m s}^{-1} \cos 20 = 6.11 \text{ m s}^{-1}$ ; horizontal distance  $s = ut = 6.11 \text{ m s}^{-1} \times 0.45 \text{ s} = 2.7 \text{ m}$

**2** Physics · May/June 2017 · Paper 01 · Q6

C (1)

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- (a) (5)
- Use of efficiency =  $\frac{\text{useful power output}}{\text{total power input}}$  (1)
  - Use of  $\Delta E_{\text{grav}} = mg\Delta h$  (1)
  - Use of  $E_k = \frac{1}{2}mv^2$  (1)
  - Use of (output) power =  $\frac{\Delta E_{\text{grav}}}{1\text{ s}} + \frac{E_k}{1\text{ s}}$  (1)
  - $v = 7.8\text{ m s}^{-1}$  (1)

Example: Output power =  $0.76 \times 160\text{ W} = 121.6\text{ W}$ ;  $121.6 = (3.5)(9.81)(0.45) + \frac{1}{2}(3.5)v^2$ ;  
 $v = 7.78\text{ m s}^{-1}$

- (b) (4)
- Use of trig to determine the vertical or horizontal component of the initial velocity (1)
  - Use of equation(s) of motion to determine time the water is in air (1)
  - Use of  $s = vt$  (1)
  - $s = 4.8\text{ m}$  (1)

Allow ecf from (a). (Using show that value,  $s = 5.0\text{ m}$ )

Example:  $u_h = 7.8 \sin 25 = 3.30\text{ m s}^{-1}$ ;  $u_v = 7.8 \cos 25 = 7.07\text{ m s}^{-1}$ ;  $0 = 7.07t + \frac{1}{2}(-9.81)t^2$ ,  $t = 1.44\text{ s}$ ;  $s = 1.44 \times 3.30 = 4.75\text{ m}$

- (c) (2)
- There is friction between the water and the pipes (1)
  - This will reduce the (initial) velocity of the water (1)

MP2 is dependent on MP1.

Use of  $v = s/t$  for the horizontal motion, or see  $u_h = 25\text{ m s}^{-1}$  (1) (4)

Use of  $s = ut + \frac{1}{2}at^2$  with  $s = 0$ , or  $v = u + at$  with  $t = 1.0\text{ s}$ , or see  $u_v = 9.81\text{ m s}^{-1}$  (1)

Combining of horizontal velocity and vertical velocity expressions, or see  $\tan \theta = \frac{9.81}{25}$  (1)

$\theta = 21^\circ$  (1)

Example of calculation:  $u_h = \frac{50\text{ m}}{2.0\text{ s}} = 25\text{ m s}^{-1}$ ;  $0 = u_v(2.0\text{ s}) + \frac{1}{2}(-9.81\text{ m s}^{-2})(2.0\text{ s})^2$ ,  $u_v = 9.81\text{ m s}^{-1}$ ;  $\tan \theta = \frac{9.81}{25}$ ,  $\theta = 21.4^\circ$